

ShadowTrace AI – UI/UX Design System

Product: ShadowTrace AI

Component: Frontend User Interface

Audience: FIGMA Designers & UX Specialists

Version: 1.0

Last Updated: September 2026

Table of Contents

1. [Design Philosophy](#)
 2. [Information Architecture](#)
 3. [Layout System](#)
 4. [Visual Design System](#)
 5. [Component Library](#)
 6. [Page Specifications](#)
 7. [State Patterns & Interactions](#)
 8. [Accessibility Guidelines](#)
 9. [Responsive Design](#)
 10. [Figma Setup Recommendations](#)
-

Design Philosophy

Core Principles

1. Trust Over Flash

- Every confidence score must be clickable to its supporting evidence
- Visual hierarchy prioritizes clarity over aesthetics
- No decorative elements that reduce data density
- Micro-interactions serve information purposes only

2. Hypothesis Language

- Always use probabilistic language ("potential correlation," "evidence suggests")
- Never use absolute language ("confirmed," "matched," "found")
- Language reflects uncertainty inherent in the system
- Copy is as important as visual design

3. Progressive Disclosure

- Dashboard shows summaries → click to expand details
- Details panel opens on demand, never auto-expand
- Multi-Level information hierarchy prevents cognitive overload
- Always show breadcrumb or back navigation

4. Dark-First & Data-Dense

- Primary theme: Dark (slate-900 background, slate-100 text)
- Optimize for long analysis sessions → reduce eye strain
- High information density prioritized over whitespace
- Data visualizations occupy maximum canvas space

5. No Client-Side Authority

- Frontend displays backend data only—never computes confidence scores
- All numbers, classifications, and statuses are backend-rendered
- Frontend enforces validation rules locally (UX only)

- Server is source of truth for all security/audit operations

Information Architecture

Site Map

```
Authentication Layer
├── /login
[Public]
├── /register
[Public, if enabled]

Main Application (requires auth)
├── /dashboard
[Analyst, Admin]
|
├── /investigations
[Analyst, Admin]
|   ├── /investigations/new
|   └── /investigations/:id
|       ├── /investigations/:id/graph
[Graph visualization]
|       ├── /investigations/:id/timeline
[Temporal analysis]
|       └──
/investigations/:id/evidence/:evidenceId
|       └──
/investigations/:id/clusters/:clusterId
|       ├── /investigations/:id/clear-web
[Clear-web resolution]
|       └── /investigations/:id/report
[Report generation]
|
├── /entities/:id
[Standalone entity view]
|
```

```
| — /settings
[User]
|   | — /settings/account                                [All
users]
|   | — /settings/fusion-config
[Admin/Analyst-lead only]
|     | — /settings/rules
[Admin/Analyst-lead only]
|
| — /audit-log
[Admin only]
```

Navigation Hierarchy

Three-Tier Navigation:

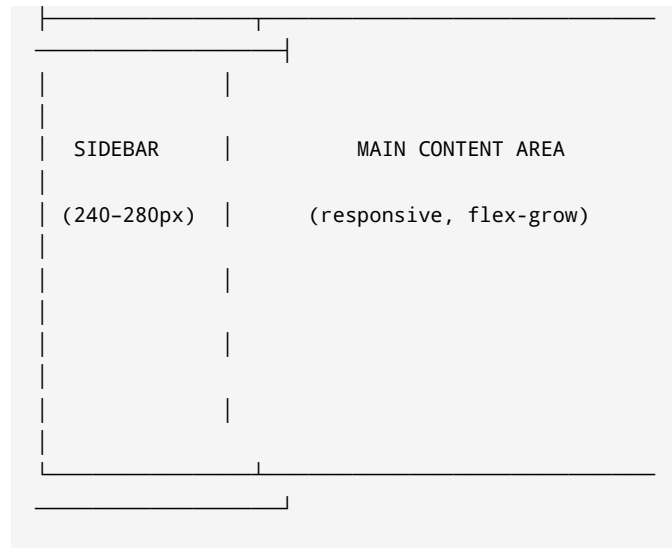
1. **Navbar** – Global: Logo, Investigations, Search, Notifications, User Avatar
2. **Sidebar** – Context-specific: varies by page (investigation tabs or dashboard filters)
3. **Page Content** – Main working area

Layout System

Master Layout Template

All authenticated pages follow this structure:

The diagram illustrates a persistent navbar structure. It features a horizontal line at the top. Below this line, a vertical line descends and then turns right to form a horizontal line. This horizontal line is labeled "Navbar (persistent)". Below this line, there are four vertical lines, each followed by a label: "Logo", "Nav", "Global Search", and "Notifications". Below the "Avatar" label, there is a vertical line.



Spacing & Grid

- **Base unit:** 8px (all spacing is multiples of 8px)
- **Grid:** 12-column layout for main content
- **Breakpoints:**
 - Desktop: 1440px+
 - Tablet: 768px-1439px
 - Mobile: < 768px

Sidebar Behavior

Page	Sidebar Content
Dashboard	Filter panel (status, date range, tags) + Recent activity
Investigation Details	Sub-navigation tabs (Overview, Graph, Timeline, Evidence, Clusters, Clear-Web, Report)

Page	Sidebar Content
Settings	Settings menu (Account, Fusion Config, Rules)
Audit Log	Filter panel + column toggles

Visual Design System

Color Palette

Confidence Score Bands (primary system)

Band	Range	Name	Color	Hex	Usage
1	0-29	Very Weak	Gray	#6B7280	Dismissed edges, weak correlations
2	30-49	Low	Blue	#3B82F6	Weak correlations, informational
3	50-69	Moderate	Amber	#F59E0B	Candidate connections, review needed
4	70-84	High	Orange	#F97316	Strong evidence, investigate further

Band	Range	Name	Color	Hex	Usage
5	85-100	Very High	Red	#EF4444	Critical evidence, urgent review

Color Philosophy: Red represents urgency and importance in investigation context, **NOT** success. A high confidence score means "look closer," not "problem solved."

Base Colors (Dark Theme)

Background:		
Surface-900:	#0F172A	(page background)
Surface-800:	#1E293B	(cards, modals)
Surface-700:	#334155	(hover states)
Text:		
Text-100:	#F1F5F9	(primary text)
Text-200:	#CBD5E1	(secondary text)
Text-400:	#94A3B8	(tertiary, disabled)
Borders:		
Border-700:	#334155	(subtle borders)
Border-600:	#475569	(strong borders)
Status Colors:		
Success:	#10B981	(confirmation, valid)
Warning:	#F59E0B	(alerts, caution)
Error:	#EF4444	(failures, critical)
Info:	#3B82F6	(information, notices)

Light Theme (Optional)

Background:

- Surface-50: #F8FAFC (page background)
- Surface-100: #F1F5F9 (cards, modals)
- Surface-200: #E2E8F0 (hover states)

Text:

- Text-900: #0F172A (primary text)
- Text-800: #1E293B (secondary text)
- Text-600: #475569 (tertiary, disabled)

Typography

Element	Font	Size	Weight	Line-Height	Usage
Heading 1	Inter	32px	700	1.2	Page titles
Heading 2	Inter	24px	600	1.3	Section headers
Heading 3	Inter	18px	600	1.4	Sub-headers
Body Large	Inter	16px	400	1.6	Primary content
Body Regular	Inter	14px	400	1.5	Standard text
Body Small	Inter	12px	400	1.4	Secondary text, labels

Element	Font	Size	Weight	Line-Height	Usage
Monospace	JetBrains Mono	13px	400	1.5	Hashes, IDs, wallet addresses

Font Stacks:

UI Font: -apple-system, BlinkMacSystemFont, 'Segoe UI', Roboto, 'Helvetica Neue', Arial, sans-serif
 Monospace: 'JetBrains Mono', 'Monaco', 'Courier New', monospace

Iconography

- **Icon Set:** Lucide Icons (or Heroicons)
- **Size:** 16px (small), 20px (standard), 24px (large)
- **Stroke Width:** 2px (consistent)
- **Color:** Inherit text color by default; colored per status

Common Icons:

- **Search:** search
- **Menu:** menu / x (close)
- **Settings:** settings
- **Alert/Warning:** alert-triangle
- **Info:** info
- **Check/Confirm:** check-circle
- **Error:** x-circle
- **Evidence:** database

- **Graph:** network
 - **Timeline:** calendar
 - **Download:** download
 - **Edit:** pencil
 - **Delete:** trash-2
-

Component Library

Form Components

Input Field

Layout:

- └─ Label (Body Small, text-200)
- └─ Input box (40px height)
 - └─ Placeholder text (text-400)
- └─ Helper text (optional, 12px, text-400)

States:

- Default (border-700)
- Focus (border-600, shadow: 0 0 0 2px rgba(59, 130, 246, 0.1))
- Error (border-red-500, error text below)
- Disabled (bg-surface-700, text-400)

Example:

Email Address	
you@example.com	
Your primary contact email	

Password Field

Layout:

- └─ Label
- └─ Input (with toggle icon for show/hide)
- └─ Strength meter (visual bar below input)

Strength Levels:

- Weak (red)
- Fair (amber)
- Good (blue)
- Strong (green)

Checkbox

Size: 18x18px

Focus: 4px border (blue ring)

States: checked, unchecked, indeterminate, disabled

Label: 14px, click-selectable (expand hit area)

Radio Button

Size: 18x18px (outer circle)

Filled dot: 8px

Focus: 4px border (blue ring)

Label: 14px, click-selectable

Spacing: 8px between options

Select/Dropdown

Trigger:

- └─ Selected value (Body Regular)
- └─ Chevron icon (right side)

└─ Height: 40px

Dropdown Menu:

- └─ Max height: 300px (scroll if needed)
- └─ Option padding: 8px 12px
- └─ Hover: bg-surface-700
- └─ Selected: checkmark icon + highlight
- └─ Dividers between groups (optional)

Toggle Switch

Layout:

- └─ Label (left side, 14px)
- └─ Toggle (40px wide × 24px tall)

States:

- Off: bg-surface-700, circle position: left
- On: bg-blue-500, circle position: right
- Focus: 2px ring
- Disabled: opacity 0.5

Button System

Primary Button

Appearance:

Background: #3B82F6 (blue)
Text: white, 14px, semi-bold
Padding: 10px 16px
Height: 40px
Border-radius: 6px

States:

- Default: bg-blue-500
- Hover: bg-blue-600

- Active: bg-blue-700
- Disabled: bg-surface-700, text-text-400, cursor-not-allowed
- Loading: spinner icon, disabled state

Secondary Button

Appearance:

Background: transparent
Border: 1px, border-600
Text: text-100, 14px
Padding: 10px 16px
Height: 40px

States:

- Default: border-600, text-100
- Hover: bg-surface-700
- Active: bg-surface-600
- Disabled: border-surface-700, text-400

Danger Button

Appearance:

Background: #EF4444 (red)
Text: white
Same sizing as Primary

Usage: Delete, archive, destructive actions

Ghost Button

Appearance:

Background: transparent
Border: none

Text: text-100 → text-blue-400 on hover
Padding: 8px 12px

Usage: Tertiary actions, links

Icon Button

Appearance:
Size: 40px × 40px (square)
Icon: 20px, centered
Border-radius: 6px

States:

- Default: transparent
- Hover: bg-surface-700
- Active: bg-surface-600
- Disabled: opacity 0.5

Focus: 2px ring on focus

Cards & Containers

Investigation Card (Dashboard)

Title	Badge
Subtitle / Description	
Entities: 12	Evidence: 45
Highest Score: 87 [Very High]	
Last Updated: 2 days ago	

Dimensions:

- Min-width: 280px

- Padding: 16px
- Gap between cards: 16px
- Grid: 3-4 columns (responsive)

States:

- Default: bg-surface-800, border-surface-700
- Hover: bg-surface-700, cursor-pointer

Evidence Card

Rule: R010 v1.2	[Type Badge]	
Confidence: 78 [High]		
Source: Blockchain Analysis		
Date Observed: Sep 20, 2026		
Details: wallet address match		

Size: Compact (fit in lists) or Full-width (detail view)

Spacing: 12px padding internal

Status Badge

Layout:

- └─ Background: color-specific
- └─ Icon (left, 16px)
- └─ Label (12px, semi-bold)
- └─ Height: 24px

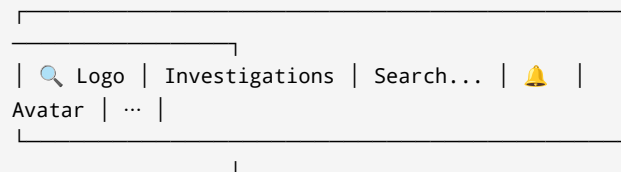
States:

- Open: blue background, circle icon
- Under Review: amber background, hourglass icon

- Closed: gray background, check-circle icon
- Archived: dimmed gray

Navigation & Structure

Navbar



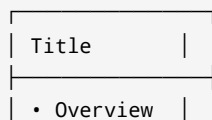
Composition:

- Left: Logo (clickable → /dashboard)
- Center-left: Nav items (Investigations)
- Center: Global search bar (250px wide)
- Right: Notification bell, User avatar, Menu
- Height: 56px
- Background: surface-800
- Border-bottom: 1px, border-700

Search Bar:

- Placeholder: "Search entities..."
- Icon: search (left side)
- Auto-complete results: dropdown below
- Enter key: navigate to entity or investigation

Sidebar (Investigation Details)



• Graph
• Timeline
• Evidence
• Clusters
• Clear-Web
• Report
Notes 

- Width: 240-280px (collapsible on tablet/mobile)
- Tab styling:
- Active: bg-surface-700, left border (3px, blue)
 - Hover: bg-surface-700
 - Icon: 20px, margin-right: 8px

Breadcrumb Navigation

Investigation › Evidence › Evidence ID #12345

- Styling:
- Font: Body Small (12px)
 - Separator: / (gray, margin: 0 4px)
 - Last item: not clickable, text-100
 - Previous items: clickable, text-blue-400
 - Use: all detail pages

Data Visualization Components

Confidence Score Display

Inline Display:

87	[Very High]
	(optional bar)

Full Component:

- Number (24px, bold)
- Band label (14px)
- Confidence bar (visual indicator)
- Tooltip on hover:
 - "Config Version: v2.1 | Last Calibrated:

Sep 15"

- Icon on outdated: ⚠

Colors: Use confidence band colors (0-29: gray, etc.)

Graph Node Legend

Legend	
• USERNAME	
◆ WALLET	
○ DOMAIN	
◀ PHONE	
◀ EMAIL	
▪ OTHER	
Confidence Bands	
— Very Weak	
— Low	
— Moderate	
— High	
— Very High	

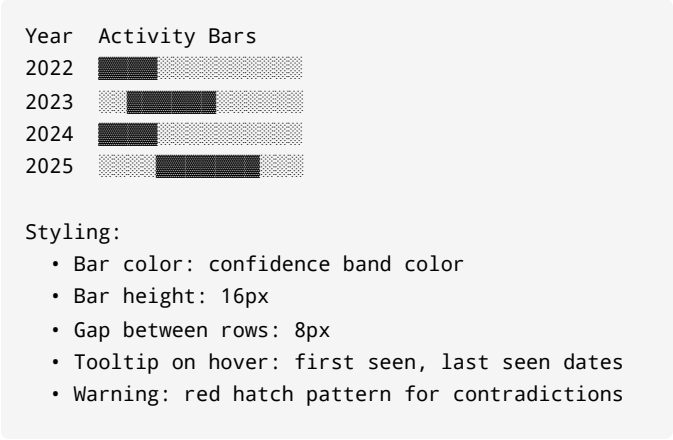
Position: Top-right of graph

Background: surface-800, 70% opacity

Padding: 12px

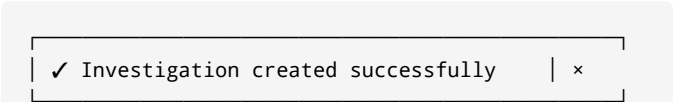
```
Border: 1px, border-600
Z-index: above graph
```

TimeLine Component



Alert & Message Components

Toast Notification



Positioning: Top-right corner

Auto-dismiss: 4 seconds

Types:

- Success: green background, check icon
- Error: red background, x-circle icon
- Warning: amber background, alert-triangle
- Info: blue background, info icon

Stack behavior: Multiple toasts stack vertically

Alert Banner

 This cluster has unresolved contradictions.
Review the evidence before finalizing.

Positioning: Full-width, top of relevant section
Background: warning-colored with opacity
Padding: 12px 16px
Border-left: 4px, warning color
Dismissible: optional × button (right side)

Empty State


No investigations yet

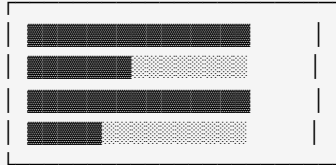
Start your first investigation
[Create Investigation Button]

Styling:

- Icon: 64px, text-400
- Heading: 18px, text-200
- Description: 14px, text-400
- Centered in container
- Padding: 48px 24px

Loading States

Skeleton Loader (Cards/Tables):



Gray pulse animation (0-0.5 opacity)
Match target element height/width
Use on cards, tables, text blocks

Spinner (Full-page transitions):

Animated circular progress indicator
Size: 32px
Color: blue-500
Position: Centered on screen
Duration: 1s rotation
Use: Page navigation, large data load

Page Specifications

Authentication Pages

Login Page (/login)

Layout:



ShadowTrace Logo
(centered, 64px)

"Investigations Start
with Evidence"

Email
[input field]

Password
[input field] [👁]

[Sign In] button

"Don't have an account?
Sign up" (link)

Width: 400px (centered)

Background: surface-900

Validation:

- Email: Real-time (local), server on submit
- Password: Field required
- Error: Generic "Invalid email or password"
(never specify which field is wrong)

Form Submit:

- On success: Set session cookie (server-side)
- On failure: Show error toast, keep form filled
- Redirect to /dashboard on success

Register Page (/register)

Similar layout to login, with:

- Email field
- Password field (with strength meter)
- Confirm password field
- "Sign Up" button
- Link to /login

Strength Meter:

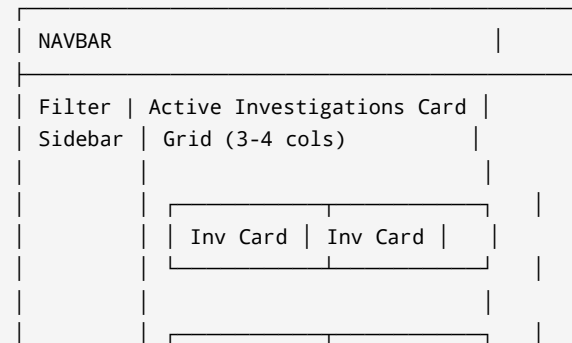
- └ Weak (< 8 chars): red bar
- └ Fair (8-11 chars + uppercase): orange bar
- └ Good (12+ chars + uppercase + number): blue bar
- └ Strong (12+ chars + upper + number + symbol): green bar
- └ Text feedback below input

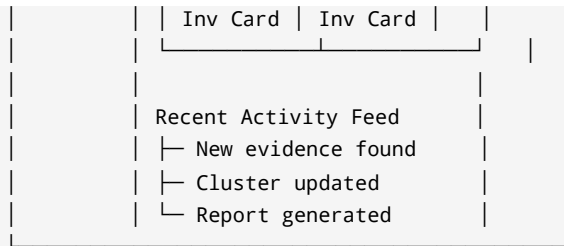
Validation Notes:

- Client-side hint only
- Server enforces: Argon2, policy checks
- Confirm password must match
- On success: Redirect to /login with success toast

Dashboard (/dashboard)

Full Page Layout:





Sidebar Filter Sections:

- Status (multi-select)
 - Open
 - Under Review
 - Closed
- Date Range
 - [From] ← → [To]
- Tags (text input, multi-select)

Investigation Card Layout:

- Title (16px, bold)
- Description (12px, text-400)
- Stats: Entities, Evidence, Highest Score
- Status badge (top-right)
- Last Updated timestamp (12px, text-400)
- On hover: Cursor pointer, bg-surface-700
- Click: Navigate to `/investigations/:id`

Recent Activity Feed:

- Timeline format (left border accent)
- Icon + action text
- Timestamp (gray, right-aligned)
- Max 5 items displayed
- Link to full audit log

CTA Button:

- "New Investigation" (top-right, primary button)
- Click: Navigate to `/investigations/new`

Empty State (if no investigations):

- Large icon ()

- Heading: "Start your first investigation"
- Description: "Create a new investigation to begin analyzing entities and relationships"
- Primary button: "Create Investigation"

Investigation List (/investigations)

Table View:

Columns: Name Created Status Highest Score				
	Investigation 1	Sep 20	Open	87 (High)
☐	Investigation 2	Sep 18	Under	64
	(Moderate)			
☐	Investigation 3	Sep 15	Closed	92 (V. High)
☐	Investigation 4	Sep 12	Open	45 (Low)
☐				

Features:

- Column headers: clickable to sort
- Row click: navigate to detail page
- Hover row: highlight bg-surface-700
- Checkbox (first column): select for bulk actions

Filters (Sidebar):

- Status: multi-select
- Date range: from/to
- Tags: multi-select
- Confidence range: slider

Bulk Actions (top, when rows selected):

- Archive selected
- Export as JSON
- "Delete" (danger action, requires confirm)

Pagination:

- 25 items per page
- Controls: ← Prev | Page 1-N | Next →
- Jump to page: text input

New Investigation (/investigations/new)

Form Layout:

New Investigation
Title *
[text input - 60 chars max]
Description
[textarea - 200 chars max]
Seed Entity (optional)
[text input - username/wallet/domain]
Help: Enter a username, wallet address, domain, or PGP fingerprint to start the investigation from a known point.
[Create Investigation] [Cancel]

Behavior:

- Title required
- Seed entity optional (will be created as first entity if provided)
- On submit: POST /api/v1/investigations

- On success: Redirect to /investigations/:id
- On error: Show error toast, keep form filled
- Cancel: Redirect to /investigations

Investigation Details Overview (/investigations/:id)

Header (persistent across all tabs):

Investigation Title	[Status ▼] 🔔	
	[Generate Report]	

Sidebar:

Overview ← active
Graph
Timeline
Evidence
Clusters
Clear-Web
Report

Main Content - Summary Cards (2 cols):

Total Entities	Total Evidence
42	127

Clusters Found	Highest Score
6	92 (Very High)

AI Summary Section:

Investigation Summary	
[AI Generated Tag]	

Prose summary of findings (2-3 paragraphs generated by backend)	
"Based on temporal analysis and clustering results..."	
[Refresh] [Copy] [Report]	

Clusters List:
 Heading: "Identified Clusters (6)"
 Grid (3 cols):

Cluster #1	
Members: 7	
Confidence: High	
⚠ Low density	
[View Details]	

[Similar cards for other clusters]

- Notes Panel (accessible from header):
- Opens as slide-over on right
 - Analyst can write/edit notes
 - Auto-saved on blur
 - Timestamps for edits

Graph View (/investigations/:id/graph)

Full Canvas Layout:
TOOLBAR (top)
[Filters ▼] [Min Conf ▼] [Layout ▼] [Weak ○]

Legend Cytoscape.js Graph Canvas

- Node
- USERNAME (circle, colored)
 - ◆ WALLET (diamond)

- Types:
- DOMAIN (hexagon)
 - Edges (lines, thickness/color)
 - ◆ by confidence band
 - Node animation on hover

◀

◀ Right Inspector Panel:

Conf [Node/Edge Title]

Bands

— Type: USERNAME

— Confidence: 87 (High)

— Observations: 12

—

— Connected to:

- @user2 (92 - V. High)
 - domain.com (64 - Moderate)
- [View All] [Explain]

Features:

- Draggable nodes
- Zoom/pan on canvas
- Click node: Inspector panel opens (right)
- Click edge: Edge inspector opens (right)
- Double-click node: Expand subgraph
- Right-click: Context menu (isolate, hide, export)

Toolbar Dropdowns:

Filter by Type: [All] ☒ USERNAME ☒ WALLET ☐ DOMAIN ☐ PHONE

Filter by Conf: [All] ☐ Very Weak ☐ Low ☒ Moderate ☒ High ☒ V.High

Layout: [Force-Directed] [Circular] [Hierarchical] [Cose]

Weak Edges: [Toggle] (hide/show < 30 confidence edges)

Inspector Panel (Right Side):

Width: 320px
Resizable (drag left edge)

For Nodes:

- └─ Entity name (monospace)
- └─ Type badge
- └─ Confidence score (with band color)
- └─ Entity ID (monospace, clickable → /entities/:id)
- └─ Observations list (scroll, collapse/expand)
- └─ Related edges list (clickable)

For Edges:

- └─ Source → Target (monospace)
- └─ Confidence score (with band color)
- └─ Rule: [Rule ID v Version]
- └─ Evidence type
- └─ Base weight / Effective weight

- └ Source observations (with dates)
- └ Integrity flags (if any, with ⚠ icons)
- └ [Explain this edge] button

Edge Explanation:

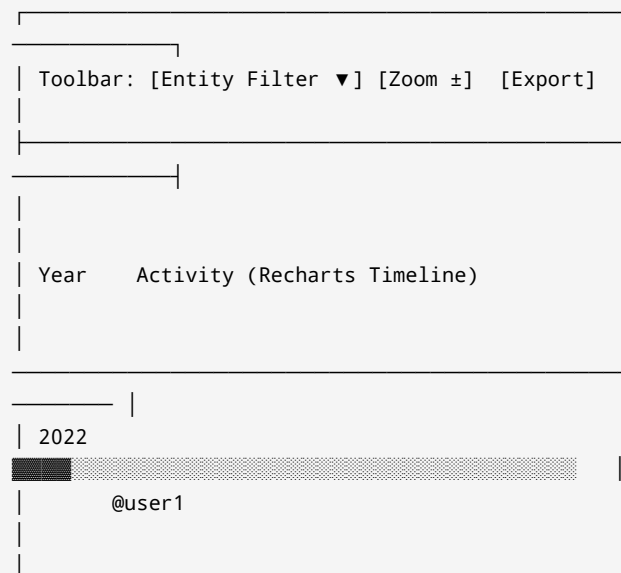
- Clicking [Explain] calls backend AI endpoint
- Shows reasoning in place of edge details
- Marked with [AI Generated] tag
- "Refresh Explanation" option
- Can be copied/exported

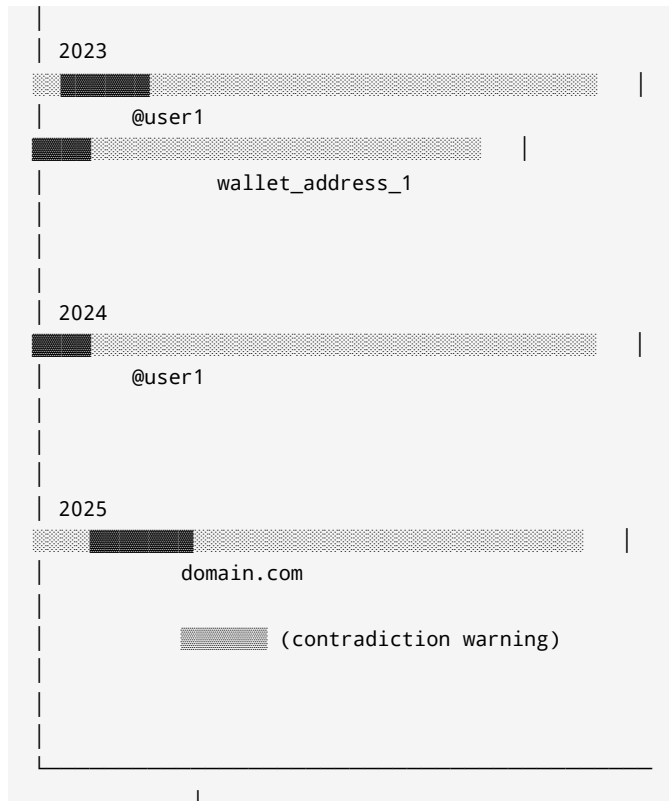
Progress Indicator:

- Shows while large cluster is laying out
- "Laying out cluster... (45%)"
- Position: Center of canvas
- Semi-transparent dark background

Timeline View (/investigations/:id/timeline)

Full Page:





Bar Styling:

- Color: Confidence band color
- Height: 16px per entity
- Opacity: 0.8
- Hover: tooltip with "First Seen: [Date] – Last Seen: [Date]"

Contradictions:

- Rendered with red hatch pattern (diagonal lines)
- Tooltip: "Temporal contradiction detected: [explanation]"
- Clickable: Links to contradiction details in Evidence section

Features:

- Zoom X-axis: Mousewheel or toolbar buttons
- Drag to pan left/right
- Filter by entity type (dropdown in toolbar)
- Export as image (PNG) or CSV

Evidence Detail

(/investigations/:id/evidence/:evidenceId)

Full Page Layout:

Breadcrumb: Investigation › Evidence › ID #12345	
Evidence Summary Card:	
@user1	↔ wallet_address_1
Confidence: 87 [Very High] (87/100)	
Rule: R010 v1.2 Type: Wallet Transfer	
Base Weight: 0.85 Effective Weight: 0.92	
Sources:	

└ Blockchain Analysis (2024-09-15)

└ Transaction Pattern (2024-09-20)

Integrity Flags:

⚠ Shared ephemeral infrastructure
(Discounted: -15% confidence)

Provenance Chain:

Observation 1: wallet_tx_001

└ Source: Blockchain Analysis

└ Date: Sep 15, 2024

└ Value: 2.5 BTC transferred

└ Independence: Group A (Dependent)

Observation 2: wallet_pattern_001

└ Source: Transaction Pattern (AI)

└─ Date: Sep 20, 2024

└─ Value: Pattern similarity 92%

└─ Independence: Group A (Dependent)

Fusion: R010 v1.2

Inputs: [wallet_tx_001, wallet_pattern_001]

Logic: weighted average of observations

Output: 0.87 confidence

Config Version: v2.1

Adjudication (Feedback Loop):

Was this evidence useful?

[✓ Confirmed Useful] [x False Positive]

Optional note:

```
| | [textarea for analyst feedback]
| |
| |
| |
| | [Submit Feedback]
| |
| |
| |
| |
| |
|_|
|_|
|_|
```

Key Features:

- Monospace for all IDs, hashes, wallet addresses
- Adjudication buttons: Prominent, easy to click
- Feedback note: Optional (sent to calibration loop)
- All AI-generated scores/summaries: marked with tag
- Integrity flags: Always visible, not hidden

Actions (top-right):

- [Share] - Copy link
- [Export] - Download as JSON
- [Back] - Return to investigation

Cluster Detail (/investigations/:id/clusters/:clusterId)

Full Page:

```
|
| Cluster #3 (7 members) [confidence band color]
|
|
```

 WARNING: Unresolved Contradictions (2)

[\[View Details\]](#) [\[Resolve\]](#)

Members List:

Entity	Confidence	Type
...		
@user1	92 (V.High)	USERNAME
@user_alt	88 (V.High)	USERNAME
wallet_0x123...	85 (V.High)	WALLET
domain.onion	72 (High)	DOMAIN
192.168.1.100	58 (Moderate)	IP
user@email.com	64 (Moderate)	EMAIL
phone:+1234567890	51 (Moderate)	PHONE

Internal Relationships Graph:

| (Filtered view of main graph – only this cluster) |

| Height: 400px

| [Toolbar same as graph view]

| Integrity Flags:

| |  Low density cluster

| | Suggestion: Review members for false merges

| |  Partial bridge audit (2 nodes)

| | Note: These nodes lack audit evidence

| | [Split Cluster] [Dismiss Flags]

| Actions (top-right):

| [Export Cluster] [View Evidence] [Merge]

Member Confidence:

- Confidence to cluster center (calculated metric)
- Use band colors for visualization
- Sortable by column

Split Cluster Action:

- Opens modal with member list
- Analyst can deselect members to keep together
- Submit creates new cluster with removed members
- Shows success toast with link to new cluster

Merge Action:

- Only if another low-confidence cluster exists
- Modal: Select cluster to merge with
- Confirmation required

Clear-Web Candidates (/investigations/:id/clear-web)

Full Page:

Clear-Web Resolution	
Cluster Confidence: 87 (Very High) – meets threshold	
Potential real-world matches identified below.	
Candidate 1:	

[Potential Clear-Web Match] (89 confidence)

Source: https://example.com/user/john_smith

Matched Features:

[Alias Match] [Email Match] [Phone Match]

Profile Info:

Name: John Smith

Email: john@example.com (matches @user1)

Phone: +1-234-567-8900

Created: 2021-05-10

[View on Web] [Confirm Match] [False Positive]

Candidate 2:

```
| | [Potential Clear-Web Match] (71 confidence)
| |
| | ... similar structure
| |
```

```
| |
| |
| | [Load More Candidates]
| |
```

Label Enforcement:

- Label is ALWAYS "Potential Clear-Web Match"
- Pulled from backend enum (never re-worded)
- Never shows "confirmed" or "identified"

Empty State (cluster too low confidence):

```
| |
| |
| | 🚫 Cluster confidence too low
| |
| | This cluster must reach 80+ confidence
| | before clear-web resolution is available.
| |
| | Current: 62 (Moderate)
| |
| | [Back to Cluster]
```

Candidate Actions:

- [View on Web]: Opens URL in new tab
- [Confirm Match]: Adjudication (→ calibration)

```
loop)
```

- [False Positive]: Adjudication with feedback

Report Generation (/investigations/:id/report)

Page Layout:

Investigation Report

[PDF Download] [JSON Export]

REPORT PREVIEW (scrollable):

ShadowTrace Investigation Report

Investigation: "Operation Nexus"

Generated: Sep 25, 2026

Analyst: John Analyst

EXECUTIVE SUMMARY

This investigation identified 6 potential clusters of related entities, with 87 evidence edges indicating correlation...

[AI Generated]

KEY FINDINGS

- └ Cluster #1: 7 entities, 92 confidence
- └ Cluster #2: 5 entities, 78 confidence
- └ Cluster #3: 4 entities, 64 confidence

TIMELINE OF ACTIVITY

(Timeline chart embedded)

- 2022: Initial activity observed
- 2023: Increased transaction volume
- 2024: Cross-platform correlation

EVIDENCE SUMMARY

- Total Evidence Edges: 127
- Very High Confidence: 34 edges (27%)
- High Confidence: 42 edges (33%)

Moderate Confidence: 38 edges (30%)

Low/Weak: 13 edges (10%)

DETAILED EVIDENCE

(Evidence cards with full details)

Rule: R010 v1.2 | Confidence: 87

Source 1: Blockchain Analysis (2024-09-15)

Source 2: Transaction Pattern (2024-09-20)

... (more evidence cards)

INTEGRITY & LIMITATIONS

[AI Generated] This report contains AI-generated summaries and analyses. All conclusions should be verified independently before use in formal proceedings.

ANALYST NOTES

"This investigation warrants further analysis of

the clear-web candidates before closure..."

[Print] [Download PDF] [Share Link]

Report Generation Flow:

1. Analyst clicks [Generate Report] from investigation
2. Backend compiles all data, runs AI summary
3. Modal shows: "Generating report... (60%)"
4. On completion: Display in page or auto-download
5. PDF includes: header, TOC, all sections, footer with timestamp

Export Options:

- PDF (formatted, ready to print/email)
- JSON (raw data, structured for external tools)

Every AI Section:

- Marked with [AI Generated] tag
- Disclaimer: "AI-generated – verify independently"
- Refresh option (re-generate summary)

Entity Detail (/entities/:id)

Full Page (Standalone):

Breadcrumb: Entities > @user1

Entity: @user1

Type: [USERNAME Badge]

First Seen: 2022-03-15

Last Seen: 2024-09-20

Observations: 23 total

Cross-Investigation Appearances:

└─ Operation Nexus (Investigation #1): 12 obs.

└─ Case Alpha (Investigation #2): 8 obs.

└─ Case Beta (Investigation #3): 3 obs.

Timeline of Activity (embedded Timeline chart):

2022: 

2023: 

2024: 

Observation History:

- Observation 1: @user1_bio_001

Source: Twitter API

Date: 2024-09-20

Value: "Bio changed to include crypto reference"
- Observation 2: @user1_follow_001

Source: Twitter Graph

Date: 2024-09-15

Value: "Followed 5 new blockchain accounts"
- [Load More] (paginated)

Related Entities (across all investigations):

[Entity chips with links, colored by type]

[Back to Investigation] [Add to New Investigation]

Use Cases:

- Clicked from graph view
- Searched directly via global search
- Cross-case correlation analysis
- Analyst reviewing entity history across cases

Settings – Fusion Config (/settings/fusion-config)

Restricted: Admin / Analyst-lead only

Full Page:

Settings › Fusion Configuration

Active Configuration: v2.1

Last Updated: Sep 15, 2026 by Admin

(Read-only view below)

Score Bands:

	Very Weak	0-29	Gray
	Low	30-49	Blue
	Moderate	50-69	Amber
	High	70-84	Orange
	Very High	85-100	Red

	Fusion Parameters:		
	Alpha (decay factor): 0.85		
	Cluster threshold: 0.70		
	Clear-web threshold: 0.80		
	Max edges per entity: 50		
	Rule Weights (sample):		
	R001 (Alias match): 0.75		
	R002 (Temporal overlap): 0.65		
	R010 (Wallet transfer): 0.85		
	...		

[Propose New Configuration]

Opens form to suggest new config (never edits in place). New config versioned as v2.2, etc.

Form includes:

└─ Diff preview (old vs new)

└─ Justification textarea

└─ [Submit for Review] [Cancel]

On submit: Notification to other admins,

Config held in "pending review" state

Previous Configurations:

v2.0 (Sep 1, 2026) - [View Diff] [Revert]

v1.9 (Aug 15, 2026) - [View Diff]

Key Points:

- Append-only versioning (never edit in-place)
- Diff view shows before/after values
- All changes require justification
- Rollback to previous config available

Settings – Rules (/settings/rules)

Restricted: Admin / Analyst-lead only

Full Page:

Settings › Rule Engine

[Add Rule] [Import Rules] [Export]

Rules Table:

ID	Version	Source → Target	Base	☐
----	---------	-----------------	------	-----------------------

R001	1.0	USERNAME-USERNAME	0.70	☐
R002	1.1	WALLET-WALLET	0.65	☒
R010	1.2	USERNAME-WALLET	0.85	☒
R015	2.0	DOMAIN-DOMAIN	0.55	☐
...				

Column Headings: Clickable to sort

Last Column (○/●): Toggle rule enabled/disabled

Row Click: Opens rule detail editor

Rule Editor (modal/side panel):

Rule ID: R001

Current Version: 1.0

→ Creating new version: 1.1

Source Entity Type: USERNAME

Target Entity Type: USERNAME

Evidence Type: Alias Match

Base Weight: 0.70

Description: Entities with matching alias
patterns are likely correlated

Logic (JSON, read-only display):

```
{ "type": "alias_similarity",  
  "threshold": 0.80, ... }
```

Enabled: ☒

[Save as v1.1] [Cancel] [Delete Rule]

Note: Saving creates new version (never edits
in place). Old versions remain in history.

Key Points:

- Editing a rule creates new version (R001 v1.1)
- Toggle enabled/disabled without versioning
- Delete (archive) available only for unused rules
- Version history shows all prior versions

Audit Log (/audit-log)

Admin-only Page

Full Page:

Settings › Audit Log

Filters (Sidebar):

Date Range: [From] - [To]

Actor: [text search]

Action: [dropdown - filter by type]

Target: [text search]

[Clear Filters] [Export]

Audit Table:

	Timestamp	Actor	Action	Target	Flags
	Sep 25 14:32	admin@...	CREATE	Inv#123	
	Sep 25 14:10	analyst@..	UPDAT	Ev#456	⚠
	Sep 25 13:45	admin@...	DELETE	Rule#1	
	Sep 24 16:20	admin@...	CONFIG	v2.1	
	...				

Click row: Expand detail (full action payload)

Pagination: 50 items per page

Action Types Displayed:

CREATE, UPDATE, DELETE, LOGIN, LOGOUT,
GENERATE_REPORT,
ADJUDICATE_EVIDENCE, SPLIT_CLUSTER,

```
CONFIG_UPDATE,  
  RULE_UPDATE, EXPORT
```

Flags Column:

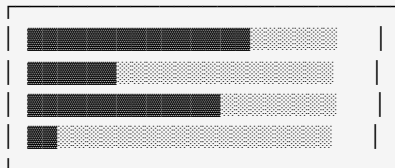
- 🚨 = security or anomaly flag
- Shows details on hover (e.g., "Mass deletion")
- Color: warning (amber)

Use: Compliance, audit trail, forensics

🧠 State Patterns & Interactions

Loading States

Skeleton Loader (Cards, tables):



Animation: Pulse (0.5-1s opacity change)

Use: When data is loading from API

Spinner (Full-page transitions):

○ ← rotating

Displayed during major page navigations

Size: 40px

Color: blue-500

Error States

Toast Notification (top-right):

```
| × Failed to create investigation |  
| Please try again. [Retry] [Dismiss] |
```

Generic Error Message (server 5xx):

"Something went wrong. Our team has been notified."

Auth Error (401):

Auto-redirect to /login, show:

"Your session has expired. Please log in again."

Validation Error (form submission):

Inline messages below input fields:

"Email format is invalid"

"Password must be at least 8 characters"

Form Interactions

Input Validation Feedback:

- Client-side: Real-time feedback (Zod)
- Server-side: Final validation
- Error message: Below input (red text, 12px)
- Success indicator: Green checkmark (optional)

Input Focus:

- Border color: Blue
- Shadow: 0 0 0 2px rgba(59, 130, 246, 0.1)
- Placeholder text fades slightly

Button Loading State:

Before: [Create Investigation]

After: [⌛ Creating...] (disabled, cursor-wait)

Form Submission:

- On error: Keep form filled, show error toast

- On success: Clear form, show success toast, navigate

Progressive Disclosure

Graph View Click Node:

- Inspector panel slides in (right, 300ms animation)
- Node is highlighted (glow effect)
- Related edges also highlight

Timeline Hover:

- Tooltip appears: "First: Sep 15 | Last: Sep 20"
- Connected nodes subtly glow

Evidence Card Expand:

- Click → Expands to show full provenance
- Animation: Smooth height increase
- Details fade in over 200ms

Accessibility Guidelines

Color & Contrast

WCAG AA Compliance:

- Minimum contrast ratio: 4.5:1 (text)
- 3:1 for large text (18px+)
- Both light and dark themes must meet this

Color Not Alone:

- Confidence bands use COLOR + TEXT LABEL
- Status badges: Icon + Text + Color
- Example: "87 (Very High)  " – red bar alone ≠ enough

Dark Theme Contrast:

- Text-100 (#F1F5F9) on Surface-900 (#0F172A):
16.5:1 ✓
- Text-200 (#CBD5E1) on Surface-800 (#1E293B):
11.2:1 ✓

Light Theme Contrast:

- Text-900 (#0F172A) on Surface-50 (#F8FAFC):
16.5:1 ✓
- Text-600 (#475569) on Surface-100 (#F1F5F9):
5.1:1 ✓

Keyboard Navigation

Tab Order:

- Navbar left-to-right
- Then main content
- Sidebar (when visible)
- Focus order matches visual flow

Keyboard Shortcuts:

- Tab: Move focus
- Shift + Tab: Move focus backward
- Enter: Activate button, submit form
- Space: Toggle checkbox, activate button
- Escape: Close modal, close sidebar panel
- Arrow Keys: Navigate within graph (optional, complex)

Focus Indicator:

- 2px outline ring (blue, all states)
- Visible on all interactive elements
- Not hidden on any element

Screen Readers

ARIA Labels:

- Icon buttons: `aria-label="Explain this edge"`
- Close buttons: `aria-label="Close panel"`
- Status badges: `aria-label="Investigation status: Open"`

Graph Canvas Fallback:

- Graph view includes list view toggle
- "This graph view is not accessible. Switch to list view?"
- List shows nodes and edges in table format

Data Tables:

- Proper `<table>` semantics
- `<th>` for headers with `scope="col"`
- Row headers for multi-column headers

Semantic HTML:

- `<button>` for buttons (not `div`s with click handlers)
- `<a>` for links (not buttons)
- `<form>` for forms
- Headings: `<h1>`, `<h2>`, etc. in order

Form Labels:

- Every input has `<label>` with `for="id"`
- Or `aria-label` if visual label is hidden

Font & Readability

Line Height:

- Body text: 1.5–1.6 (160%)
- Headings: 1.2–1.4 (120–140%)
- Line length: 60–80 characters max (optimal)

Font Size Minimum:

- Body text: 14px (12px only for

secondary/tertiary)

- Buttons/inputs: 14px minimum
- No text smaller than 10px

Letter Spacing:

- Normal (0) for body text
- Can increase by 0.5-2% for better readability
- Avoid kerning reduction

Font Weights:

- 400 (normal) for body
- 600 (semi-bold) for headings
- 700 (bold) for emphasis
- Avoid weights < 400

Responsive Design

Breakpoints & Strategy

Desktop (1440px+):

- Sidebar always visible (240px)
- 3-4 column grid for cards
- Full graph canvas
- All features enabled

Tablet (768px-1439px):

- Sidebar collapses to hamburger icon (top-left)
- 2 column grid for cards
- Graph view pinch-to-zoom enabled
- Inspector panel narrows to 250px

Mobile (< 768px):

- Sidebar hidden, hamburger menu
- Single column layout (100% width)
- Read-only mode (no editing/adjudication)

- Graph disabled: "Full editing available on desktop"
- Inline message on graph view

Desktop-First Approach:

Design for desktop first, then progressively simplify
for smaller screens. Don't create mobile-first and scale up.

Responsive Components

Navigation:

Desktop: Horizontal navbar + sidebar
Tablet: Hamburger menu (opens sidebar slide-over)
Mobile: Hamburger menu, sidebar slides in from left

Cards:

Desktop: 3 columns, 280px min-width
Tablet: 2 columns, 300px min-width
Mobile: 1 column, 100% width - padding

Inputs:

Desktop: Full width or constrained (max 500px)
Tablet: Full width
Mobile: Full width, larger padding for touch

Tables:

Desktop: Full table view
Tablet: Horizontal scroll enabled
Mobile: Stack view or card view (not table)

Modals:

Desktop: Center, 600px width, 80vh max-height
Tablet: Center, 90% width, 90vh max-height
Mobile: Full screen, edge-to-edge

Touch Targets

Minimum tap target: 44px × 44px (mobile)
Desktop click target: 32px × 32px minimum

Button Sizing:

- Height: 40px (standard), 48px (mobile-specific)
- Padding: 10px 16px (desktop), 12px 18px (mobile)
- Spacing between interactive elements: ≥ 8px

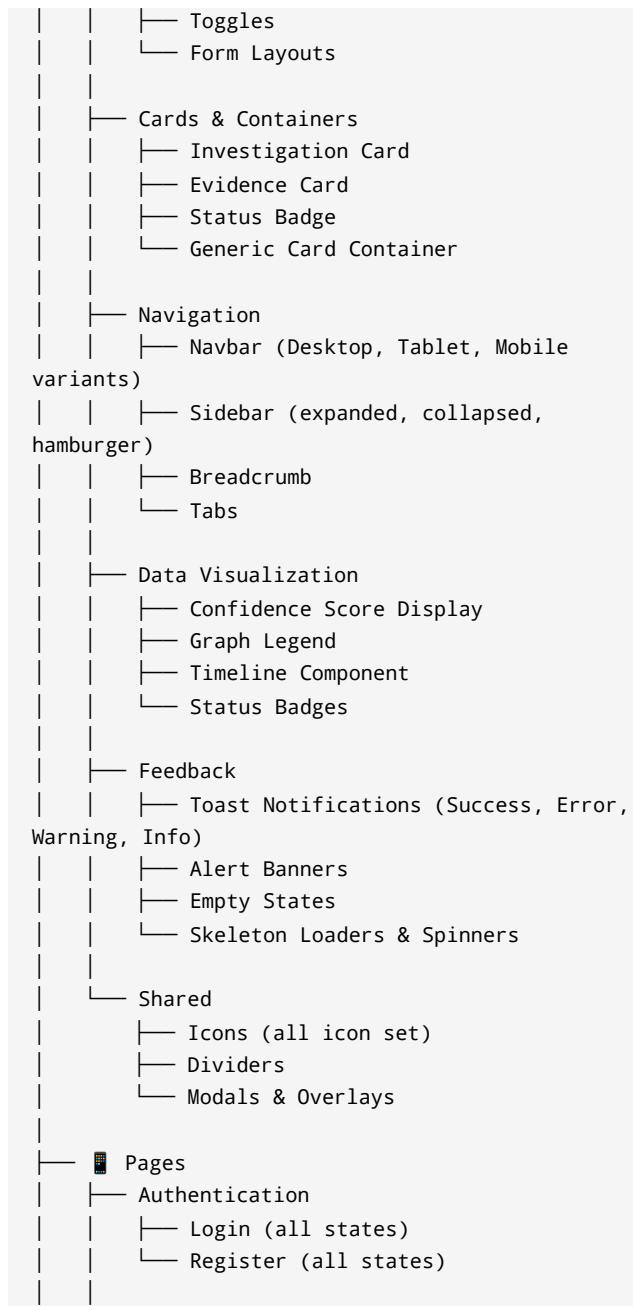
Icon Buttons:

- 40px × 40px (desktop)
- 48px × 48px (mobile)
- Icon size: 20px (desktop), 24px (mobile)

Figma Setup Recommendations

File Structure

```
ShadowTrace_UI_Design
├── Foundation
│   ├── Color Palette
│   ├── Typography System
│   ├── Grid & Spacing
│   └── Breakpoints
├── Components
│   ├── Form Components
│   │   ├── Input Fields
│   │   └── Buttons (Primary, Secondary, Danger, Ghost, Icon)
│   ├── Checkboxes & Radio Buttons
│   └── Selects & Dropdowns
```



- |
 - |
 - | Dashboard
 - | Full Layout (with active items)
 - | Empty State
 - | Loading State
 - | Investigations
 - | Investigation List (full table + mobile)
 - | New Investigation (form layout)
 - | Investigation Details
 - | Overview
 - | Graph View (with inspector)
 - | Timeline View
 - | Evidence Detail
 - | Cluster Detail
 - | Clear-Web Candidates
 - | Report
 - | Entity Detail
 - | Settings
 - | Account Settings
 - | Fusion Configuration
 - | Rules
 - | Audit Log
 - | Responsive Variants
 - | Desktop (1440px+)
 - | Tablet (768-1439px)
 - | Mobile (< 768px)
 - | 📱 Prototypes
 - | User Flow: Login → Dashboard → Investigation
 - | User Flow: Create Investigation
 - | User Flow: Adjudicate Evidence
 - | Interaction: Graph Node Click → Inspector Panel

Figma Features to Use

Components & Variants:

- Create component for Button with variants: type (Primary, Secondary, Danger, Ghost), size (small, medium, Large), state (default, hover, active, disabled, loading)
- Create component for Badge with variants: status (open, under-review, closed), color (confidence bands)
- Use component swap to quickly change button styles

Constraints:

- Use "scale" for elements that should resize with frame
- Use "left/right/center" for fixed positioning
- Set up proper constraints for responsive behavior

Auto Layout:

- Use for consistent spacing on all components
- Set spacing, alignment, and padding rules
- Cards, buttons, and sections should use auto layout

Grids & Guides:

- Set up 8px grid system (base unit)
- Create column guides (12 columns) for page layouts
- Use component grid for icon consistency

Text Styles:

- Create library of text styles: H1, H2, H3, Body Large, Body Regular, Body Small, Mono
- Link to typography system file for consistency

Color Styles:

- Create color styles for: Surface colors, text colors, confidence bands
- Name systematically: Color/Surface/900 ,
Color/Text/100 , Color/Confidence/VeryHigh

Prototyping:

- Link pages together for user flow testing
- Add interactions: click button → show panel (slide animation)
- Test navigation between major pages

Design Tokens (Recommended Export)

```
{
  "colors": {
    "surface": {
      "900": "#0F172A",
      "800": "#1E293B",
      "700": "#334155",
      "600": "#475569"
    },
    "text": {
      "100": "#F1F5F9",
      "200": "#CBD5E1",
      "400": "#94A3B8"
    },
    "confidence": {
      "veryWeak": "#6B7280",
      "low": "#3B82F6",
      "moderate": "#F59E0B",
      "high": "#F97316",
      "veryHigh": "#EF4444"
    }
  },
  "typography": {
    "h1": { "fontSize": "32px", "fontWeight": 700,
    "lineHeight": 1.2 },
    "body": { "fontSize": "14px", "fontWeight":
```

```
400, "lineHeight": 1.5 },
  "mono": { "fontSize": "13px", "fontFamily":
"JetBrains Mono", "lineHeight": 1.5 }
},
"spacing": {
  "xs": "4px",
  "sm": "8px",
  "md": "16px",
  "lg": "24px",
  "xl": "32px"
}
}
```

Summary

This UI/UX design system provides:

- ✓ **Comprehensive specs** for every page and component
- ✓ **Dark-first design** optimized for analyst workflows
- ✓ **Progressive disclosure** to prevent information overload
- ✓ **Accessibility-first** (WCAG AA compliant)
- ✓ **Responsive design** from desktop to mobile
- ✓ **Trust-focused visual language** (no false certainty)
- ✓ **Figma-ready** component structure and organization

Next Steps for Implementation:

1. **Import this document into Figma** as design spec reference
2. **Build component library** following the structure outlined
3. **Create page templates** for each route
4. **Set up design tokens** and export for frontend team
5. **Create prototypes** for key user flows
6. **Test accessibility** with screen readers and keyboard navigation

7. Iterate based on developer feedback during handoff

Version History:

- v1.0 (Sep 25, 2026) – Initial comprehensive design system

Contact: Design System Team | **Last Updated:** Sep 25, 2026